# Muaaz Bhamjee

Associate Professor, Department of Mechanical and Aeronautical Engineering
University of Pretoria

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Language: English Gender: Male Citizenship: South African

## ABOUT ME

Prof. Muaaz Bhamjee is currently an Associate Professor in the Department of Mechanical and Aeronautical Engineering at the University of Pretoria. He is the Founder and Principal Investigator of the CERG-FLUX Lab (Fluids, Learning and Uncertainty in compleX systems). He has a background in Mechanical Engineering and Applied Mathematics. He is involved in research and has published in the fields of Computational Fluid Dynamics (CFD), Experimental High Energy Physics (HEP), Heat and Mass Transfer, Experimental Fluid Dynamics, Lattice Boltzmann Modelling, Multiphase Flow, Positron Emission Particle Tracking (PEPT), High Performance Computing (HPC), Solar and Renewable Energy, Solar Air Heating, Atomic Layer Deposition (ALD) Modelling, Fluid-Structure Interaction (FSI), Fluid Dynamics in Biomedical Applications and Engineering Education. The primary focus of his research was in the use of Navier-Stokes and Lattice-Boltzmann based computational fluid dynamics (CFD) approaches in modelling of the multi-scale fluid dynamics, heat and mass transfer processes and AI and data-driven approaches to Climate. Currently he is involved in research related to data driven and machine learning approaches to fluid dynamics, high energy physics, dynamical systems and hybrid quantum computing and machine learning approaches to such.

Prof. Bhamjee's prior experience includes roles as Staff Research Scientist at IBM Research – Africa, Senior Research Associate (Visiting) at University of Johannesburg, Senior Lecturer (and prior Lecturer) at University of Johannesburg as well as Junior Computational Fluid Dynamics (CFD) Analyst at Hatch Africa.

## RESEARCH LAB LEADERSHIP

Founder and Principal Investigator, CERG-FLUX Lab (Fluids, Learning and Uncertainty in compleX systems) (2026–present)
Research subgroup within the Clean Energy Research Group (CERG)
Department of Mechanical and Aeronautical Engineering, University of Pretoria

## PROFESSIONAL PROFILES

| | |
|---|---|
| **LinkedIn** | linkedin.com/in/muaaz-bhamjee-ding-mechanical-preng-000b7212 |
| **UP Profile** | up.ac.za – Staff Profile |
| **SAAM ExCo** | saam.africa/exco |

## RESEARCH PROFILES

| Profile | ID | Link | H-index |
|---|---|---|---|
| Google Scholar | Prof. Muaaz Bhamjee | Link | 68 |
| ORCID | 0000-0002-2697-4589 | Link | N/A |
| Scopus | 55247084100 | Link | 37 |
| ResearchGate | Muaaz Bhamjee | Link | 37 |
| Web of Science | ABE-4708-2020 | Link | 32 |

H-indices as at 27 October 2026; all include ATLAS Collaboration papers.

## FIELDS OF SPECIALISATION

- Computational Fluid Dynamics (CFD).

- Mechanics (Statics and Dynamics).
- Heat and mass transfer.
- Multiphase flow.
- Navier-Stokes and Lattice Boltzmann based computational methods for modelling fluid flow, heat and mass transfer.
- Numerical analysis.
- Data Science, Machine Learning, Deep Learning.
- Geospatial Data Analysis and Climate Sciences.
- High Performance Computing (HPC) – application of HPC to modelling.
- Solar and Renewable Energy / Solar Air Heating.
- Positron Emission Particle Tracking (PEPT): application and source preparation.
- Course co-ordination and lecturing.
- Scholarship of teaching and learning: Engineering education research with a focus on design literacy, online and blended learning.

## PROFESSIONAL SKILLS

- Research and Patent Ideation.
- Significant experience in research environments in academia and in industry.
- Journal and conference paper reviewing.
- Post-graduate supervision.
- Mentoring of Interns and Students.
- Project Management and Leadership.
- Course co-ordination and lecturing; Programme co-ordination.
- Supervision of final year students for their capstone research and design projects.
- Familiar with the engineering consulting and process engineering environments.
- Technical report writing; Teaching academic writing.
- Interpersonal skills; Teamwork and co-ordination; Mentoring.

## TECHNICAL SKILLS

- Proficient in Python for data science, machine and deep learning and solving of mathematical problems and general programming.
- Experience with Python-based Machine and Deep Learning Frameworks: PyTorch, TensorFlow and Keras, as well as scikit-learn.
- Geospatial and Climate Data Analysis.
- Computer Science and Software Engineering Skills: DevOps, Agile Development and Scrum, Cloud Computing, Unix/Linux and Shell Scripting, Git and GitHub, Containerisation of applications/software, Kubernetes and OpenShift.
- Proficient with ANSYS Fluent for CFD.
- Proficient with Palabos – open-source libraries for Lattice Boltzmann based approach to CFD.
- Proficient in HPC hardware selection and application of HPC.
- Proficient with Matlab, Simulink, Maxima, GNU Plot, Excel and C++ for solving of mathematical problems and general programming.
- Proficient in LaTeX for general and mathematical typesetting.
- Proficient with Microsoft Office and OpenOffice.

## TERTIARY EDUCATION

**Doctor of Engineering (DIng Mechanical Engineering Science)**
*University of Johannesburg* 2011–2016 (examined 2016; conferred 2017)
*Thesis:* Modelling of the Multiphase Interactions in a Hydrocyclone using Navier-Stokes and Lattice Boltzmann based Computational Approaches.
Supervisor: Prof. S.H. Connell (Department of Physics, UJ)
Co-supervisor: Prof. A.L. Nel (Department of Mechanical Engineering Science, UJ)
Link to Thesis

**Master of Engineering (MIng Mechanical Engineering Science) – *Cum Laude***
*University of Johannesburg* 2009–2011
*Dissertation:* A Computational Fluid Dynamics and Experimental Investigation of an Airflow Window.
Supervisor: Prof. A. Nurick (Department of Mechanical Engineering Science, UJ)
Co-supervisor: Mr. D.M. Madyira (Department of Mechanical Engineering Science, UJ)
Link to Dissertation

**BSc Honours (Applied Mathematics) – *Cum Laude***
*University of Johannesburg* 2008–2010
*Minor Dissertation:* The Link between the Finite Element and Finite Difference Methods for the Poisson Equation.
Supervisor: Dr. J. Prentice (Department of Applied Mathematics, UJ)

**Bachelor of Engineering (BIng Mechanical Engineering Science)**
*University of Johannesburg* 2004–2008

## SECONDARY EDUCATION

**RAUCALL Secondary School** 1999–2003
Highest Grade Passed: Grade 12 (Matric)
Subjects: English (First Language) HG, Afrikaans (Second Language) HG, Mathematics HG, Accounting HG, Physical Science HG, Economics HG.

## EMPLOYMENT HISTORY

**Associate Professor** January 2025 – Present
*University of Pretoria*
Line Manager: Prof. Schalk Kok
Co-ordination and lecturing of the third year Impact of Engineering Activity and Groupwork (MIA 320) module. Lecturing of the Scientific Machine Learning component of MKM 411 (self-developed ANN and PINN material). Lecturing of the self-developed design thinking and design process unit in JSU 120 (since 2025). Supervision of final year mechanical engineering students' research and design projects. Supervision of post-graduate students. Research.

**Staff Research Scientist** February 2023 – December 2024
*IBM Research – Africa*
Line Managers: Dr. Sibusisiwe A. Makhanya and Dr. Etienne Vos
Conduct Research within the IBM Research Accelerated Discovery Pillar with specific focus on data-driven and machine learning approaches to Climate. Challenge/Project Lead, Patent Ideation, Intern and Student Mentorship.

**Senior Research Associate (Visiting)** July 2023 – December 2024
*University of Johannesburg*
Line Managers: Prof. T.C. Jen and Prof. Nkosinathi Madushele
Conduct Research and Post-graduate supervision.

**Senior Lecturer** May 2017 – January 2023
*University of Johannesburg*
Line Managers: Prof. E.T. Akinlabi and Prof. T.C. Jen
Co-ordination and lecturing of first year Introduction to Engineering Design 1A and 1B modules. Supervision of final year project investigation and design projects. Supervision/co-supervision of Masters and Doctoral students. Departmental co-ordinator for the First Year Seminar. Departmental software and computer labs co-ordinator. Teaching and learning representative on the Departmental Industry Advisory Board (IAB). Research – Computational Fluid Dynamics (CFD) and engineering education. Former co-ordinator of the Master of Sustainable Energy Programme. Former Departmental short learning programmes (SLPs) co-ordinator.

**Lecturer** January 2015 – April 2017
*University of Johannesburg*
Line Manager: Prof. E.T. Akinlabi
Co-ordination and lecturing of first year Introduction to Engineering Design 1A and 1B modules. Supervision of final year project investigation and design projects. Co-supervision of Masters students and research.

**Junior Engineer (CFD Analyst)** 2008–2009
*Hatch Africa Pty (Ltd)*
Line Manager: Mr. S. Ritchie
Computational Fluid Dynamics (CFD) modelling for the metals and mineral processing environment. Research of relevant theory, experimental procedures and material properties. Programming of User Defined Functions (C compiler) to improve CFD model accuracy. Technical report writing.

## TEACHING

### Modules Presented

| Module | Level | Self-developed |
|---|---|---|
| MIA 320: Impact of Engineering Activity and Groupwork (UP; module co-ordinator; 2025–present) | Third year | Partly: GA7 and GA8 assessment strategy, rubrics and Engineers Without Borders design challenge |
| MKM 411: Computational Fluid Dynamics, Scientific Machine Learning component (UP; 2026–present; with Prof. K.J. Craig) | Fourth year | Partly: ANN and PINN material self-developed |
| JSU 120: design thinking and design process unit (UP; 2025–present) | First year | Yes (unit self-developed) |
| Introduction to Engineering Design 1A and 1B (UJ; module co-ordinator; 2015–2023) | First year | Yes (largely redeveloped) |
| Final-year research and design project supervision (UJ 2015–2023; UP 2025–present) | Fourth year | Not applicable |

### Other Courses Presented

- Lecturer, 7th African School of Fundamental Physics and Applications (ASP2022): *Computational Fluid Dynamics (CFD) 1* and *CFD 2*, 5 December 2022. Link 1, Link 2
- Departmental short learning programmes (co-ordinator), University of Johannesburg.

### Learning Materials Developed

- MKM 411 Scientific Machine Learning: ANN and PINN lecture material, Jupyter notebooks and a local, POPIA-compliant notebook autograder (2026).
- MIA 320: study guide and module design, including an Engineers Without Borders group design challenge (GA7 and GA8).
- JSU 120: design thinking and design process unit and learning material (2025–present).
- Open-source teaching and postgraduate training repositories (2026), github.com/cerg-flux-lab: MKM411 SciML lecture notebooks; PINN heat transfer (with Prof. K.J. Craig); Introduction to LaTeX, Overleaf and Git; Introduction to Git; research proposal guide; lab paper guide; Claude guide; AI research guardrails.
- Introduction to Engineering Design 1A and 1B (UJ): largely redeveloped module content and assessment (2015–2023).

## ACADEMIC ADMINISTRATION AND CITIZENSHIP

- Departmental Marketing Co-ordinator, Department of Mechanical and Aeronautical Engineering, University of Pretoria.
- Member, Departmental Management Committee, Department of Mechanical and Aeronautical Engineering, University of Pretoria.
- Member, HPC Steering Committee (institutional level), University of Pretoria.
- EBIT Faculty representative (one of two), Future Africa strategic research group, University of Pretoria.

## AFFILIATIONS

- UP Institutional Representative/Team Leader within the ATLAS Collaboration at CERN; Principal Investigator (PI) in the SA-CERN ATLAS Project; new ATLAS Authorship Qualification Task completed in 2026 (time-dependent $N_2$ flushing simulation in the ATLAS ITk Strips).
- Principal Investigator, UP Quantum Science and Technology Hub (UPQuST), UP's node in the South African Quantum Technology Initiative (SA QuTI).
- Registered Professional Engineer (PrEng) with the Engineering Council of South Africa (ECSA).
- Full member of the South African Institute of Mechanical Engineering (SAIMechE).
- Full member of the South African Association for Theoretical and Applied Mechanics (SAAM).
- Vice-President of the SAAM Executive Committee (SAAM ExCo).
- Vice-President of the South African National IUTAM (International Union of Theoretical and Applied Mechanics) Committee.
- Former Secretary of the SAAM Executive Committee (SAAM ExCo).
- Former Secretary of the South African National IUTAM Committee.
- Former Full member of the South African Society for Engineering Education (SASEE). Membership concluded in 2023.

## COMMUNITY ENGAGEMENT AND PROFESSIONAL SERVICE

- Vice-President, SAAM Executive Committee (former Secretary).
- Vice-President, South African National IUTAM Committee (former Secretary).
- Electra Mining Africa 2026 (Nasrec, 7–11 September 2026): co-ordinated the Department of Mechanical and Aeronautical Engineering contribution to the UP EBIT presence.
- Public engagement on 702 and CapeTalk (*The Aubrey Masango Show*, "Weird and Wonderful"): interview on UP hosting UPQuST, the new national quantum technology hub (2026). Link
- Public engagement on careers in engineering and science: Radio Islam interview and *Al-Qalam* feature (2025).

## RECOGNITION

- NRF rating: Not yet rated.
- 2025 Breakthrough Prize in Fundamental Physics, awarded to the ATLAS Collaboration (co-recipient as a collaboration member).
- NRF Thuthuka Funding Instrument (Post-PhD Track), 2017–2019 (see Grants and Other Funding Awarded).

## GRANTS AND OTHER FUNDING AWARDED

- **2026 – UP Quantum Science and Technology Hub (UPQuST) SAQuTI-Linked Grant**
  *Hybrid Quantum-Classical Computing Accelerated Computational Fluid Dynamics*
  Total Value: R180,000
- **2026 – UP Quantum Science and Technology Hub (UPQuST) SAQuTI-Linked Grant**
  *Quantum-Machine Learning for Rare Particle Searches in High Energy Physics Data*
  Total Value: R180,000
- **2026 – UP Quantum Science and Technology Hub (UPQuST) SAQuTI-Linked Grant**
  *Quantum-magnetic Sensors Particle Detection in Mineral Separation*

Total Value: R400,000

- **2025 – UP Research Development Programme (RDP)**
  *Advancing Computational Techniques for Multiphase Flow: Lattice Boltzmann Method, Deep Learning and Quantum Computing Approaches*
  Total Value: R50,000
- **2022 – UJ URC Research Equipment Grant**
  *Preventing COVID-19 with Mathematical Modelling, Digital twins, AI and 4IR Engineered Solutions*
  Total Value: R284,517.15
- **2021 – UJ GES COVID-19 Grant**
  *Preventing COVID-19 with Mathematical Modelling, Digital twins, AI and 4IR Engineered Solutions*
  Total Value: R80,000
- **2020–2021 – European Synchrotron Radiation Facility (ESRF)**
  *Design and study of equipment for the BEATS beamline at SESAME and other equipment for ESRF beamlines*
  Total Value: €8,750
- **2017–2019 – NRF Thuthuka Funding Instrument (Post-PhD Track)**
  *Modelling of Multiphase Flow in Process Equipment*
  Grant Number: TTK160615171464 Total Value: R602,000

### Principal Investigator – HPC Projects

- *Multiphase Flow Modelling and Deep Learning*, Project No. MECH1739, Centre for High Performance Computing (CHPC). *Active.*
- *Modelling of Multiphase Flow in Process Equipment*, Project No. MECH0837, CHPC. Concluded 31 January 2023.
- *Preventing COVID-19 with Mathematical Modelling, Digital twins, AI and 4IR Engineered Solutions.* Concluded 31 January 2023.

## FORMAL COLLABORATIONS

- *Modelling of Multiphase Flow in Process Equipment* – Collaborator: Multotec. Period: 2017–2019. Status: Concluded.
- *Design and study of the experimental hutch equipment for the BEATS beamline at SESAME* – Collaborator: ESRF. Period: 2020–2022. Status: Concluded.
- *Optimized Heat Transfer for X-Ray Optical Components Designed for High Vibration Stability* – Collaborator: ESRF. Period: 2019–2023. Status: Involvement Ceased.

## SCHOLARSHIPS AWARDED

- January 2012 – June 2014: Department of Science and Technology (DST) Innovation Scholarship for Doctoral Study.
- July 2011 – June 2014: University of Johannesburg Next Generation Scholarship for Doctoral study.
- January–June 2011: NRF Extension Scholarship for Masters studies.
- January–March 2011: ENERKEY for Masters Study.
- 2009–2010: Department of Science and Technology (DST) Innovation Scholarship for Masters Study.
- 2009–2010: University of Johannesburg Next Generation Scholarship for Masters studies.
- 2004–2007: Group 5 (Pty) Ltd scholarship for undergraduate studies.

## POSTGRADUATE SUPERVISION

### Masters – Graduated

- T. Jooma Abbajee, “Reservoir Computing: A Tool for Predicting Chaotic Dynamical Systems”, 2026, University of Johannesburg *Co-supervisor.*
- K.M. Makhanya, “The Use of Computational Fluid Dynamics for Assessing Flow-Induced Acoustics to Diagnose Lung Conditions”, 2025, University of Johannesburg. *Co-supervisor.*
- T.F. Mokoena, “Design and study of the experimental hutch equipment for the BEATS beamline at SESAME”, University of Johannesburg, 2022. Spent six months at the ESRF. *Main Supervisor.*
- T. Ramokoka, “Modelling Fluid and Particulate Flow Through a Ventriculoperitoneal Shunt in a Variable Temperature Environment”, University of Johannesburg, 2022. *Main (Sole) Supervisor.*
- T.B. Molale, “Numerical Modelling of a Low-Pressure Steam Turbine’s Last Stage Rotating Blade”, MEng, University of Johannesburg, 2021. *Co-supervisor.*
- A. Potgieter, “Modelling of a heated gas-solid fluidised bed using Eulerian based models”, MEng, University of Johannesburg, 2020. *Cum Laude. Main Supervisor.*
- D.M. Chirnside, “A Numerical Investigation of Air-Core Formation and Particle Separation in a Hydrocyclone”, MIng, University of Johannesburg, 2019. *Main (sole) supervisor.*
- S.A.T. Tina, “Design Optimization of a Hydrocyclone: Effect of the Cone Shape on the Hydrocyclone Separation Efficiency”, MIng, University of Johannesburg, 2017. *Cum Laude. Co-supervisor.*
- M.S.W. Potgieter, “An Experimental and Computational Fluid Dynamics Investigation of a Counter/Parallel Flow Solar Air Heater”, MIng, University of Johannesburg, 2016. *Cum Laude. Co-supervisor.*

### Doctoral – Ongoing

- Maxwell Chambakata, "Lattice Boltzmann Method for Modelling Two-Phase Flow", University of Pretoria. *Main supervisor.*
- A.Y. Akinbowale, "Quantum Machine Learning for Rare Particle Signatures in High-Dimensional Physics Data", University of Pretoria. *Main supervisor.*
- M. Ramokali, "Hybrid classical-quantum computing and machine learning methods for computational fluid dynamics", University of Pretoria. *Main supervisor.*
- M. Takoutchouang Saphira Agnes, "An integrated study of fluid–structure interactions in pathological arteries: development of numerical and experimental models for understanding pathophysiological mechanisms and optimizing methods to improve blood flow accuracy", University of Pretoria. *Main supervisor.*
- M.S.W. Potgieter, "Hybrid Lattice Boltzmann - Machine Learning methods for two-phase flow", Univerisity of Pretoria. *Main supervisor.*
- A. Essop, "A novel numerical scheme for high speed flows", University of Pretoria. *Main supervisor.*
- T. Jooma Abbajee, “Graph dynamical Reservoir Computing", University of Johannesburg *Co-supervisor.*
- M.P. Connell, “Search for Higgs boson decays via dark bosons to a four-lepton final state”, University of Johannesburg. *Co-supervisor.*

### Masters – Ongoing

- M. Maila, "Modelling of multiphase flow in a hydrocyclone", University of Pretoria. *Main supervisor.*
- B. Kazembe, "Modelling of a heated gas-solid fluidised bed using an Eulerian-Eulerian granular model and a Dense Discrete Phase Model (DDPM)", University of Pretoria. *Main supervisor.*
- M.Y. Rasool, "Searches for BSM Higgs Decay Channels in the ATLAS Experiment at the Large Hadron Collider", University of Pretoria. *Co-supervisor.*

### Masters – Supervision Ceased

- Confidence Malatjie, "Computational Fluid Dynamics and Physics Informed Neural Networks", University of Johannesburg – Applied Mathematics. *Co-supervisor. Supervision Ceased February 2026.*

### Doctoral – Supervision Ceased

- I. Makani, "Characterization of hydrochar prepared from food waste for the adsorption of $H_2S$ from biogas using deep learning, experiment and techno-economic analysis", University of Pretoria. *Co-supervisor. Supervision Ceased November 2025.*
- M.T.L. Lekwana, "Computational Fluid Dynamics Study of Hemodynamics in the Human Cardiovascular System", University of Johannesburg. *Main supervisor. Supervision Ceased January 2023.*
- A. Potgieter, "Optimized Heat Transfer for X-Ray Optical Components Designed for High Vibration Stability", University of Grenoble-Alpes (UGA) / ESRF. *Co-supervisor. Supervision Ceased February 2023.*

### ATLAS Authorship Qualification Task (AQT) – Local Supervision

- M.P. Connell, "Computational Fluid Dynamics Modelling of the $N_2$ Purging Flow Field, Temperature, Humidity and Dew Point Distributions in the Inner Pixels for the ATLAS ITk Upgrade" – Completed 1 March 2022. *Local Supervisor.*
- P. Mafa Takisa, "Computational Fluid Dynamics Modelling of the Flow Field, Temperature, Humidity and Dew Point Distributions in the Outer Service Volume (OSV) for the ATLAS ITk Upgrade" – Supervision Ceased 31 January 2023. *Local Supervisor.*
- A.Y. Akinbowale, ATLAS Authorship Qualification Task: "Anomaly detection on operational time series of the ATLAS TDAQ" – Started 2026, ongoing. *Local Supervisor.*

### Mentorship

- Raees Ebrahim Moosa, IBM Research – Africa Mentorship Programme, BSc Honours Computer Science, University of Witwatersrand. Period: April 2024 – Present.
- R. Seipei, IBM Research Africa – Intern. Period: April – October 2023.
- Raees Ebrahim Moosa, IBM Research – Africa Mentorship Programme, BSc Computer Science, University of Witwatersrand. Period: April – October 2023.

## POSTGRADUATE EXAMINATION

### Doctoral

- Qudus Boluwatife Jimoh-Taiwo, "Development of a Computational COVID-19 Thrombosis Model: Understanding the Role of Flow", PhD, University of Cape Town, 2025. *External examiner.*

### Masters

- M Motlana, "Translating CFD Tools Into Clinical Practice: A Clinically Applicable CFD Pipeline for Interventional Planning in Coarctation of the Aorta", MSc in Biomedical Engineering, University of Cape Town, 2026. *External examiner.*
- TM Mashego, "Numerical modelling of a wonderstone thermal energy storage system.", MEng, University of Johannesburg, 2026. *External examiner.*
- Brenald Dzonzi, "Physics-Informed Neural Networks as Surrogates for Haemodynamics in Coarctation of the Aorta", MEng, University of Cape Town, 2025. *External examiner.*
- Sadman Sakib Ahmed, "Computational modelling of hydrogel therapies", MEng, University of Cape Town, 2025. *External examiner.*
- PB Mathiva, "Machine Learning Model Development for Predictive Maintenance in Industrial Robotics.", MEng, University of Johannesburg, 2025. *External examiner.*

- SM Khumalo, "Experimental and numerical analysis of granular material flowability and heat exchange.", MEng, University of Johannesburg, 2025. *External examiner.'*
- KT Hlabyago, "Influence of Vehicle Cabin Thermal Conditions on Short-Drive Cooling Demand and Passenger Comfort.", MEng, University of Pretoria, 2025. *Internal examiner.*
- P Jeewa, "CFD Investigation into the Role of Impingement Length and Pitch in Jet Impingement Boiling.", MEng, University of Pretoria, 2025. *Internal examiner.*
- T Sewell, "Development of a Submersible Robot Inspection Platform for Pipes", MSEN, University of Kwazulu-Natal, 2023. *External examiner.*

### Undergraduate Module Moderation

- Fourth-year Heat and Mass Transfer and Computational Fluid Dynamics modules, Emeris. *External moderator.*

## JOURNAL PUBLICATIONS

*(Accredited Journals. ATLAS/CERN collaboration papers listed below as indexed on Google Scholar.)*

### Non-ATLAS Journal Papers

1. Khanyisani Mhlangano Makhanya and Muaaz Bhamjee and Neil Martinson and Simon Connell, "Assessment and characterization of the impact of pulmonary pathology on flow-induced acoustics using computational fluid dynamics", *Biomedical Signal Processing and Control*, vol. 120, pp. 110018, 2026. DOI
2. Ralijaona, Mbolahasina and Igumbor, Emmanuel and Bhamjee, Muaaz and Otwombe, Kennedy and Nabeemeeah, Firdaus and Milovanovich, Minja and Martinson, Neil and Mafa, Pedro and Leeuw, Lerothodi and Connell, Simon, "Infectiousness model of expelled droplets exposed to ultraviolet germicidal irradiation coupled with evaporation", *Computers & Fluids*, vol. 275, pp. 106242, 2024. DOI
3. Mokoena, Fortune and Bhamjee, Muaaz and Connell, Simon H. and Van Vaerenbergh, Pierre and Iori, Gianluca and Kaprolat, Axel, "An FEA Investigation of the Vibration Response of the BEATS Detector Stage", *R&D Journal of the South African Institution of Mechanical Engineering*, vol. 39, pp. 44–52, 2023. DOI
4. Makhanya, K. M. and Connell, S. H. and Bhamjee, Muaaz and Martinson, N. A., "The Use of Computational Fluid Dynamics for Assessing Flow-Induced Acoustics to Diagnose Lung Conditions", *Mathematical and Computational Applications*, vol. 28, No. 3, pp. 64, 2023. DOI
5. Bhamjee, Muaaz and Connell, Simon H. and Nel, André Leon, "The Modification of the Dynamic Behaviour of the Cyclonic Flow in a Hydrocyclone under Surging Conditions", *Mathematical and Computational Applications*, vol. 27, No. 6, pp. 88, 2022. DOI
6. Potgieter, A. and Bhamjee, M. and Kruger, S., "Modelling of a Heated Gas-solid Fluidised Bed using Eulerian Based Models", *R&D Journal of the South African Institution of Mechanical Engineering*, vol. 37, pp. 45–57, 2021. DOI
7. Potgieter, M. S. W. and Bester, C. R. and Bhamjee, M., "Experimental and CFD Investigation of a Hybrid Solar Air Heater", *Solar Energy*, vol. 195, pp. 413–428, 2020. Impact Factor: 5.742. DOI
8. Coetzee, Rigardt Alfred Maarten and Jen, Tien-Chien and Bhamjee, Muaaz and Lu, Junling, "The mechanistic process analysis and temperature effect in a low pressure square type atomic layer deposition reactor", *International Journal of Mechanical Engineering and Robotics Research*, vol. 8, No. 5, pp. 792–796, 2019. DOI
9. Coetzee, Rigardt Alfred Maarten and Jen, Tien-Chien and Bhamjee, Muaaz and Lu, Junling, "The mechanistic effect over the substrate in a square type atomic layer deposition reactor", *International Journal of Modern Physics B*, vol. 33, No. 01n03, pp. 1940018, 2019. DOI
10. Bhamjee, M. and Nurick, A. and Madyira, D. M., "An Experimentally Validated Mathematical and CFD Model of a Supply Air Window: Forced and Natural Flow", *Energy and Buildings*, vol. 57, pp. 289–301, 2013. Impact Factor: 5.879. DOI

## ATLAS/CERN Collaboration Papers

*Note: Papers on which Bhamjee is a contributing author, as indexed on Google Scholar.*

1. ATLAS Collaboration, "Observation of pair production of longitudinally polarized $Z$ bosons in four-lepton final state with the ATLAS detector", *Physical Review Letters*, 2026. DOI
2. ATLAS Collaboration, "Search for boosted low-mass resonances decaying into hadrons produced in association with a photon in pp collisions at $\sqrt{s} = 13$ TeV with the ATLAS detector", *Journal of High Energy Physics*, vol. 2025, No. 1, pp. 99, 2025.
3. ATLAS Collaboration, "Measurements of $W^+W^-$ production cross-sections in $pp$ collisions at $\sqrt{s} = 13$ TeV with the ATLAS detector", *Journal of High Energy Physics*, vol. 2025, No. 8, pp. 1–66, 2025.
4. ATLAS Collaboration, "The performance of missing transverse momentum reconstruction and its significance with the ATLAS detector using 140 fb$^{-1}$ of $\sqrt{s} = 13$ TeV $pp$ collisions", *The European Physical Journal C*, vol. 85, No. 6, pp. 606, 2025.
5. ATLAS Collaboration, "Differential cross-section measurements of Higgs boson production in the $H \to \tau^+\tau^-$ decay channel in $pp$ collisions at $\sqrt{s} = 13$ TeV with the ATLAS detector", *Journal of High Energy Physics*, vol. 2025, No. 3, pp. 1–66, 2025.
6. ATLAS Collaboration, "Search for diphoton resonances in the 66 to 110 GeV mass range using $pp$ collisions at $\sqrt{s} = 13$ TeV with the ATLAS detector", *Journal of High Energy Physics*, vol. 2025, No. 1, pp. 53, 2025. DOI
7. ATLAS Collaboration, "Observation of $\Upsilon$ Production in Pb+Pb Collisions at $\sqrt{s_{\mathrm{NN}}}$ with the ATLAS Detector", *Physical Review Letters*, vol. 134, No. 14, 2025.
8. ATLAS Collaboration, "Exploration at the high-energy frontier: ATLAS Run 2 searches investigating the exotic jungle beyond the Standard Model", *Physics Reports*, vol. 1116, pp. 301–385, 2025.
9. ATLAS Collaboration, "The quest to discover supersymmetry at the ATLAS experiment", *Physics Reports*, vol. 1116, pp. 261–300, 2025.
10. ATLAS Collaboration, "Characterising the Higgs boson with ATLAS data from the LHC Run-2", *Physics Reports*, vol. 1116, pp. 4–56, 2025.
11. ATLAS Collaboration, "Climbing to the Top of the ATLAS 13 TeV data", *Physics Reports*, vol. 1116, pp. 127–183, 2025.
12. ATLAS Collaboration, "Software and computing for Run 3 of the ATLAS experiment at the LHC", *The European Physical Journal C*, vol. 85, No. 3, pp. 234, 2025.
13. ATLAS Collaboration, "A search for R-parity-violating supersymmetry in final states containing many jets in $pp$ collisions at $\sqrt{s} = 13$ TeV with the ATLAS detector", *Journal of High Energy Physics*, vol. 2024, No. 5, pp. 003, 2024.
14. ATLAS Collaboration, "A search for top-squark pair production, in final states containing a top quark, a charm quark and missing transverse momentum, using the 139 fb$^{-1}$ of $pp$ collision data collected by the ATLAS detector", *Journal of High Energy Physics*, vol. 2024, No. 7, pp. 1–49, 2024.
15. ATLAS Collaboration, "Combination of measurements of the top quark mass from data collected by the ATLAS and CMS experiments at $\sqrt{s} = 7$ and 8 TeV", *Physical Review Letters*, vol. 132, No. 26, pp. 261902, 2024.
16. ATLAS Collaboration, "Evidence for the $VH, H \to \tau\tau$ process with the ATLAS detector in Run 2", *Physics Letters B*, vol. 855, pp. 138817, 2024.
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## PATENTS

1. **"Dynamically Forecasting High Resolution Air Temperature in Real-Time Using Multiple Sources"**

   Inventors: Akhalwaya, Ismail Yunus and Vos, Etienne Eben and Makhanya, Sibusisiwe Audrey and Govindasamy, Tamara Rosemary and Patel, Zubeida and Morake, Dedricks Monyai and Mahlasi, Craig and Bhamjee, Muaaz.
   Inventors on behalf of IBM Corp. South Africa: Granted 30 July 2024, ZA2023/09448 (ID: 99151468). USA: Pending, US18/200,050. Link

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- *JOM – Journal of The Minerals, Metals & Materials Society (TMS)* (special issue)
- *Applied Energy* – Elsevier
- *Building and Simulation* – Springer
- Reviewer: *SAIP 2025* (69th Annual Conference); *SACAM 2025* (14th Conference)
- Reviewer: *ICLR 2024* – Three workshop proposals reviewed.
- Reviewer: *SAIP 2024* (68th Annual Conference)
- Reviewer: *WEEF/IFEES/GEDC 2022*, Cape Town (two abstracts)
- Reviewer: *SAIP 2022* (66th Annual Conference)
- Reviewer: *WEEF/GEDC 2021*, Madrid, Spain (two papers)
- Abstract reviewer: *2nd Biennial SOTL in the South Conference*, Bloemfontein, October 2019
- Reviewer: *SMPM 2019*, North West, 8–10 March 2019
- Reviewer: *SACAM 2018*, Vanderbijlpark, 17–19 September 2018
- Reviewer: *SACAM 2012*, Johannesburg, 3–5 September 2012

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- Member of the Local Organising Committee for *SACAM 2012*, Johannesburg, 3–5 September

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